

AQI ANALYTICS



Exploring Air Quality
Trends Using SQL

INDIA'S Air Quality
Landscape



Introduction

Air pollution is one of the most pressing environmental challenges in India, affecting public health, quality of life, and sustainable development. Monitoring and analyzing Air Quality Index (AQI) data is essential for understanding pollution patterns and identifying high-risk regions.

This project analyzes over 200,000+ AQI observations collected across Indian states and cities using SQL. Through data exploration, aggregation, and trend analysis, the study identifies pollution hotspots, compares air quality across regions, and highlights the most prominent pollutants contributing to poor air quality.

The objective of this analysis is to transform large-scale environmental data into meaningful insights that support data-driven decision-making and increase awareness of air quality conditions across India.

```
--- total records
```

```
SELECT COUNT(*) AS total_records  
FROM aqi_data;
```

	total_records
▶	235785

TOTAL RECORDS.

The dataset contains 235785 AQI records, providing extensive coverage of air quality observations across Indian states and cities.

A large dataset ensures reliable trend analysis, regional comparisons, and meaningful insights into air pollution patterns across India.

--- Average AQI

```
SELECT ROUND(AVG(aqi_value),2) AS average_aqi  
FROM aqi_data;
```

	average_aqi
▶	98.14
	98.14

AVERAGE AQI.

The overall average AQI across 707,355 observations was 98.14, providing a baseline measure of air quality across the dataset.

However, this average may mask significant regional variations, with some locations experiencing much higher or lower AQI levels.

AQI TREND OVER TIME

```
--- AQI Trend Over Time  
SELECT DATE_FORMAT(date, '%Y-%m') AS month_year,  
        ROUND(AVG(aqi_value),2) AS avg_aqi  
FROM aqi_data  
GROUP BY month_year  
ORDER BY month_year;
```

AQI levels consistently peaked during November–January and improved significantly during June–September, indicating a clear seasonal pattern in air quality. For example, AQI reached 164.48 in Dec 2022 but dropped to 50.30 in Jul 2022.

Air quality showed gradual improvement over the years, with peak AQI decreasing from 164.48 (Dec 2022) to 137.42 (Nov 2024) and monthly AQI remaining below 120 throughout the available 2025 data.

month_year	avg_aqi
2022-01	120.60
2022-02	102.25
2022-03	90.14
2022-04	50.30
2022-05	54.05
2022-06	58.27
2022-07	93.63
2022-08	152.25
2022-09	164.48
2022-10	161.02
2022-11	131.05
2022-12	104.42
2023-01	103.37
2023-02	93.81
2023-03	81.22
2023-04	58.11
2023-05	70.42
2023-06	63.99
2023-07	101.92
2023-08	154.15
2023-09	138.35
2023-10	143.24
2023-11	104.49
2023-12	92.75
2024-01	95.39
2024-02	92.04
2024-03	76.37
2024-04	52.68
2024-05	47.10
2024-06	53.99
2024-07	84.87
2024-08	137.42
2024-09	116.97
2024-10	115.77
2024-11	95.70
2024-12	86.76
2025-01	82.36

TOP 10 POLLUTED CITIES

```
--- Top 10 Polluted Cities
SELECT area,
       ROUND(AVG(aqi_value),2) AS avg_aqi
FROM aqi_data
GROUP BY area
ORDER BY avg_aqi DESC
LIMIT 10;
```

Byrnihat, Begusarai, and Sri Ganganagar emerged as the most polluted cities, each recording an average AQI close to or above 200.

Several cities from Bihar and northern India appear in the top 10, suggesting higher pollution levels in these regions.

area	avg_aqi
Byrnihat	240.00
Begusarai	206.68
Sri Ganganagar	197.94
Darbhanga	190.09
Dharuhera	185.72
Siwan	181.23
Hanumangarh	177.82
Chhapra	176.25
Muzaffarnagar	175.57
Samastipur	170.89

STATE WISE AQI ANALYSIS

```
--- State-wise AQI Analysis
SELECT state,
        ROUND(AVG(aqi_value),2) AS avg_aqi
FROM aqi_data
GROUP BY state
ORDER BY avg_aqi DESC;
```

Himachal Pradesh, Jharkhand, and Bihar recorded the highest average AQI values, indicating relatively poorer air quality among the analyzed states.

Telangana, Mizoram, Meghalaya, and Sikkim reported the lowest AQI levels, reflecting comparatively cleaner air conditions.

state	avg_aqi
Himachal Pradesh	160.26
Jharkhand	156.01
Bihar	141.85
Haryana	132.41
Rajasthan	120.81
Odisha	119.50
Punjab	117.85
Tripura	116.01
Assam	104.93
Delhi	101.81
Uttar Pradesh	97.21
West Bengal	94.79
Madhya Pradesh	93.88
Gujarat	90.45
Uttarakhand	87.90
Maharashtra	81.04
Chandigarh	80.68
Nagaland	80.46
Andhra Pradesh	72.02
Jammu and Kashmir	69.50
Kerala	63.59
Chhattisgarh	63.51
Manipur	62.52
Tamil Nadu	61.16
Karnataka	58.75
Andaman and Nicobar Islands	57.71
Puducherry	56.51
Arunachal Pradesh	54.49
Sikkim	53.69
Meghalaya	50.07
Mizoram	47.25
Telangana	28.52

POLLUTANT DISTRIBUTION

```
--- Pollutant Distribution
SELECT prominent_pollutants,
       COUNT(*) AS frequency
FROM aqi_data
GROUP BY prominent_pollutants
ORDER BY frequency DESC;
```

PM10 was the most frequently reported pollutant with over 111,000 observations, followed by PM2.5 with nearly 60,000 observations.

Particulate matter pollutants (PM10 and PM2.5) dominated the dataset, indicating that airborne particles are the primary contributors to air pollution.

prominent_pollutants	frequency
PM10	111053
PM2.5	59670
"PM2.5	16906
O3	16202
CO	12867
"PM10	9634
SO2	3815
NO2	3012
"O3	1020
"CO	984
"NO2	428
"SO2	109
NH3	69
"SO3	15
"NH3	1

TOP 5 CLEANEST CITIES

```
--- Top 5 Cleanest Cities
SELECT area,
       ROUND(AVG(aqi_value),2) AS avg_aqi
FROM aqi_data
GROUP BY area
ORDER BY avg_aqi ASC
LIMIT 5;
```

Bengaluru recorded the lowest average AQI of 20, making it the cleanest city in the dataset.

All top five cleanest cities maintained average AQI values below 30, indicating consistently good air quality conditions.

area	avg_aqi
Bengaluru	20.00
Chennai	20.63
Mumbai	23.73
Pune	26.23
Hyderabad	28.52

CONCLUSION

- Analysis of 235,000+ AQI observations revealed significant variations in air quality across Indian cities and states between 2022 and 2025.
- Byrnihat emerged as the most polluted city with an average AQI of 240, while Bengaluru recorded the lowest average AQI among major cities.
- PM10 and PM2.5 were the dominant pollutants, indicating that particulate matter is the primary contributor to air pollution.
- AQI levels followed a clear seasonal pattern, with higher pollution levels during winter months and improved air quality during the monsoon season.
- The dashboard provides an interactive way to monitor air quality trends, compare regions, and support data-driven environmental analysis.

A pair of hands is shown holding a small plant seedling with two leaves. The background is a dark, textured surface covered with many small, dark leaves, creating a sense of a forest floor or a garden bed. The lighting is soft, highlighting the hands and the seedling.

Air quality is not just an environmental concern but a public health challenge. Continuous monitoring and data-driven decision-making can help identify pollution hotspots and support effective pollution control measures.

~ Itishree Sahu